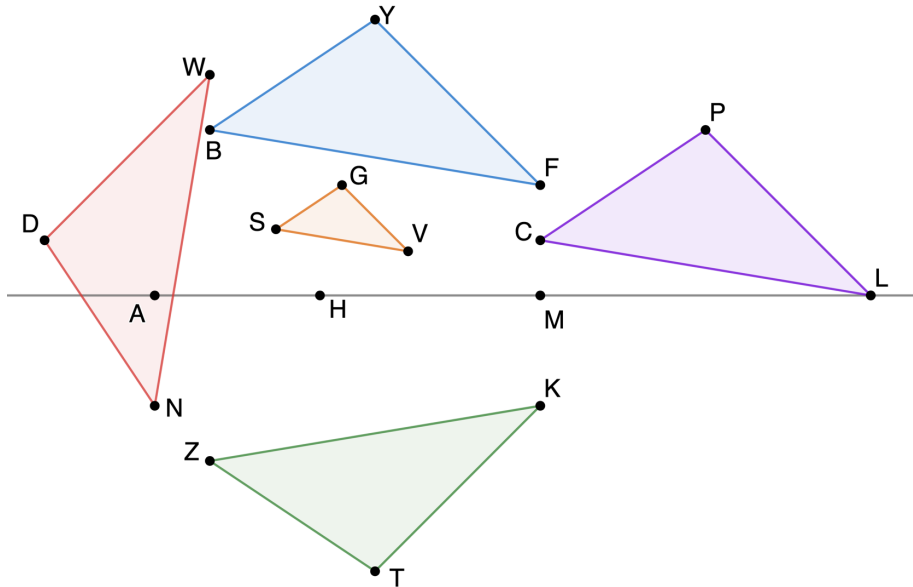




1.6.5 Wrap-Up: Ticket Out the Door

- The diagram shows four different single transformations of $\triangle YFB$.



Complete the chart by naming and describing the type of transformation that was applied to $\triangle YFB$ to result in each triangle.

Triangle	Type of Transformation	Description of the Transformation
$\triangle PLC$		
$\triangle TKZ$		
$\triangle GVS$		
$\triangle DWN$		

Unit 1, Lesson 7: Congruence and Similarity



Benchmarks

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.



Learning Targets

- ☐ I can recognize the relationship between corresponding sides and corresponding angles of congruent and similar figures.
- ☐ I can write congruence statements to relate corresponding parts following a rigid transformation.
- ☐ I can write similarity statements to relate corresponding parts following a dilation.



1.7.1 Warm-Up: 10,000 Hours to Awesome

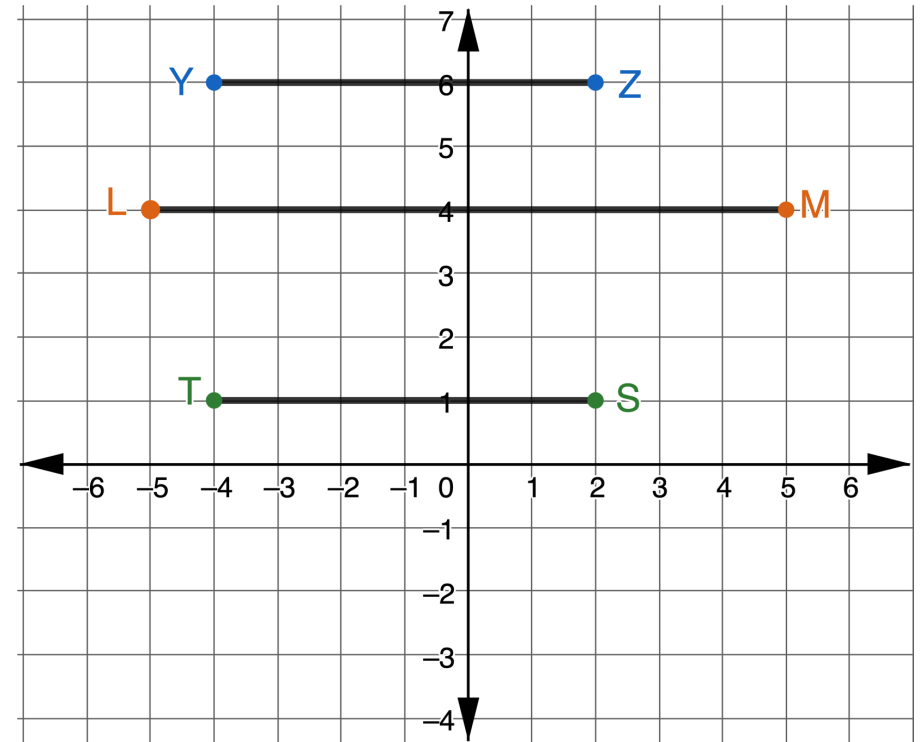
Best-selling author Malcolm Gladwell states in his book "Outliers" that it takes 10,000 hours of practice to be a phenom — so freakishly awesome or great at something. In 1973, Herbert Simon and William Chase wrote a paper published in *American Scientist*. Simon and Chase did experiments with chess-playing tasks and computer simulations to determine how long it would take someone to reach the grandmaster level in the game of chess. Their conclusion was that it could take as many as 50,000 hours staring at chess positions to become a chess grandmaster.

1. What skill, hobby, or activity would you like to become freakishly awesome at?
2. How many years is 10,000 hours?
3. How many years is 50,000 hours?



1.7.2 Collaboration: Line Segments

1. $\overline{LM}$, $\overline{TS}$, and $\overline{YZ}$ are shown on the coordinate grid.



- a. What do you notice about the three line segments?

- b. List at least two different ways to show that the length of $\overline{TS}$ is equal to the length of $\overline{YZ}$.

- c. What is the ratio of the length of $\overline{TS}$ to the length of $\overline{LM}$?

- d. What is the ratio of the length of $\overline{TS}$ to the length of $\overline{YZ}$?

- e. Describe a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{LM}$.

- f. Draw a line that extends through the coordinate plane and passes through points L and T .

- g. Draw a line that extends through the coordinate plane and passes through points M and S .

- h. Where do the two lines intersect?

- i. What is a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{YZ}$?

- j. Ask two classmates the transformation they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

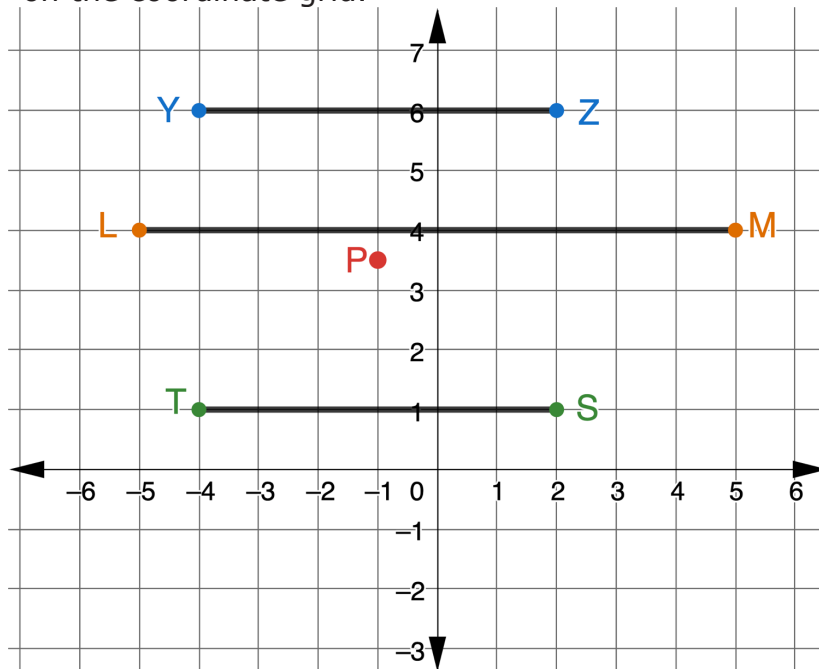
Transformation	My Summary of Their Reason	Initials



2. Day'Ron and Zeb were discussing how to transform $\overline{TS}$ to create $\overline{YZ}$.

Day'Ron	Zeb
The only two single transformations are a translation of up 5, or a reflection across the line $y = 3.5$.	There are three single transformations. A translation of up 5, a reflection across the line $y = 3.5$, and a rotation of 180° .

Day'Ron disagreed with Zeb. To prove his point, he used a graphing utility to show $\overline{TS}$, $\overline{YZ}$, $\overline{LM}$ and point P on the coordinate grid.



Day'Ron said, "It is not possible to rotate $\overline{TS}$ 180° so that point T aligns with point Y and point S aligns with point Z ."

Zeb replied, "Yes, but that does not matter because the letters do not have to match exactly. If you rotate $\overline{TS}$ 180° about point P it will coincide with $\overline{YZ}$."

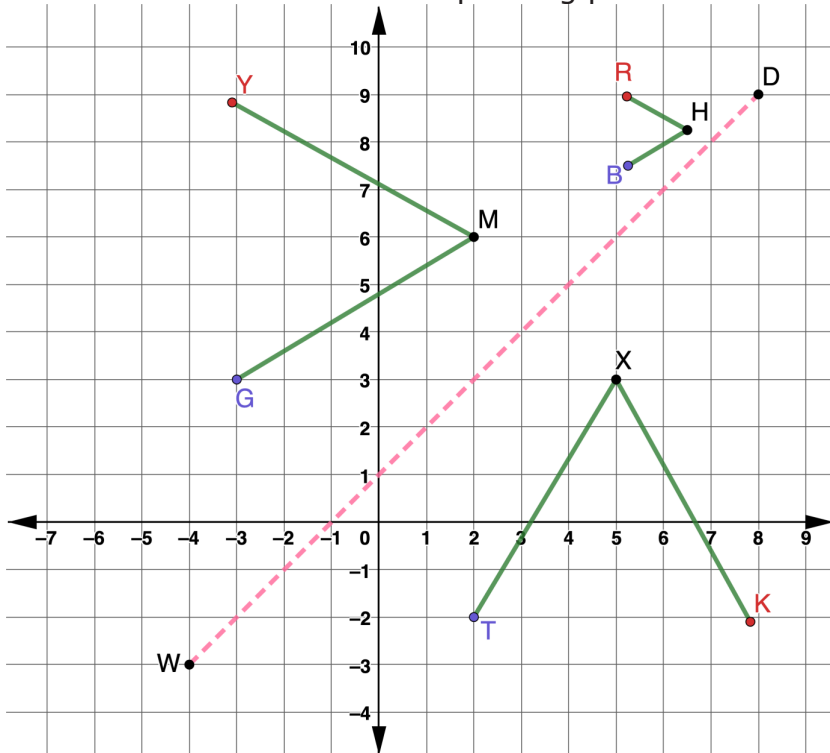
Discuss Day'Ron and Zeb's justifications with your partner. Summarize your discussion.

- Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed congruence?
- Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed similarity?



1.7.3 Guided Practice: Angles

$\overline{WD}$, $\angle YMG$, $\angle RHB$, and $\angle KXT$ are shown on the coordinate grid. Each color relates to corresponding points.



1. Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle KXT$.
2. Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle RHB$.

3. Use patty paper to create $\angle ALZ$ by rotating $\angle YMG$ 90° counterclockwise centered on point M.
4. Use patty paper to create $\angle CNF$ by translating $\angle YMG$ left 4 down 1.
5. Which angles have the same measure as $\angle YMG$?
6. Which transformation(s) resulted in an angle whose measurement is the same as $\angle YMG$?



1.7.4 Bring It Together: Congruence and Similarity

1. Complete the statements.
 - a. A rigid transformation results in a line segment that

☐ is

☐ is not

 congruent to the original line segment.
 - b. A dilation results in a line segment that

☐ is

☐ is not

 congruent to the original line segment.
 - c. A rigid transformation results in an angle that

☐ is

☐ is not

 congruent to the original angle.
 - d. A dilation results in an angle that

☐ is

☐ is not

 congruent to the original angle.



Two figures are **congruent** if they have exactly the same shape and size. Equivalently, two figures are congruent if one figure can be mapped to the other using a single rigid motion, or a sequence of transformations that includes only rigid motions.



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.

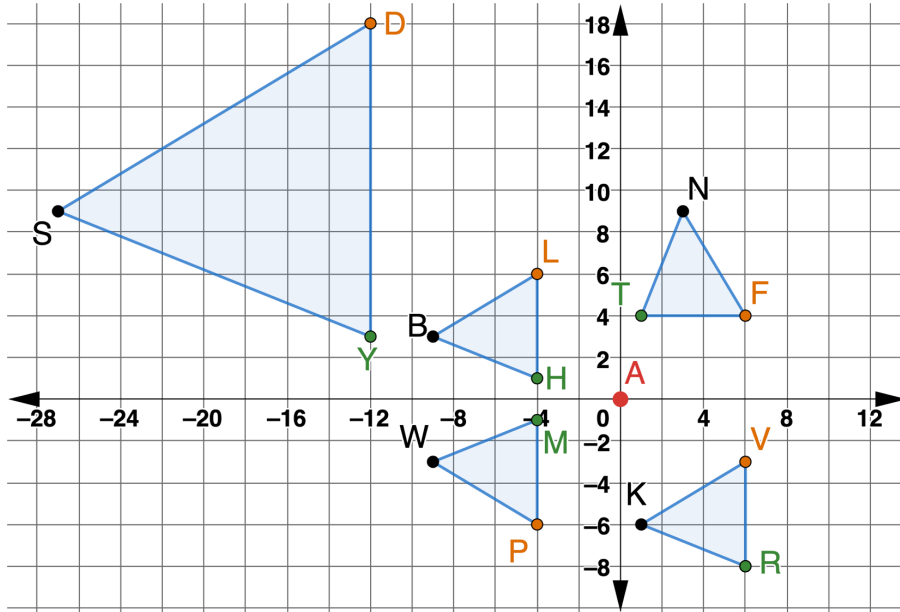
Two congruent figures can also be considered similar with a scale factor of 1. Although this is known, it is not often required to state that two congruent figures are also similar.

2. Which statement in Question 1 would be false for a dilation with a scale of factor of 1?



1.7.5 Guided Instruction: Congruence and Similarity Statements

1. Five triangles are shown on the coordinate grid. $\triangle FNT$, $\triangle VKR$, $\triangle PWM$, and $\triangle DSY$ are created using a single transformation of $\triangle LBH$. Each color relates to corresponding points.



a. Complete the table.

Triangle	Description of Transformation	Congruent or Similar
$\triangle FNT$		
$\triangle VKR$		
$\triangle PWM$		
$\triangle DSY$		

- b. Use the information about the transformations to complete each statement.

$$\overline{LH} \cong \underline{\hspace{2cm}} \quad \overline{TN} \cong \underline{\hspace{2cm}} \quad \overline{YS} \sim \underline{\hspace{2cm}}$$

$$\angle DYS \cong \underline{\hspace{2cm}} \quad \overline{MP} \sim \underline{\hspace{2cm}} \quad \underline{\hspace{2cm}} \cong \overline{LH}$$

$$\overline{PW} \cong \underline{\hspace{2cm}} \quad \underline{\hspace{2cm}} \cong \angle LHB \quad \angle VKR \cong \underline{\hspace{2cm}}$$

- c. Discuss with your partner why there could be multiple answers to each statement. Summarize your discussion.



1.7.6 Wrap-Up: Cool Down

Complete the statements about today's lesson.

1. Something I learned today ...

2. I didn't know ...

3. One question I still have ...



Unit 1, Lesson H1: Symmetries



Benchmark

MA.912.GR.2.4 Determine symmetries of reflection, symmetries of rotation, and symmetries of translation of a geometric figure.



Learning Targets

- ☐ I can determine symmetries of reflection.
- ☐ I can determine symmetries of rotation.
- ☐ I can determine symmetries of translation.